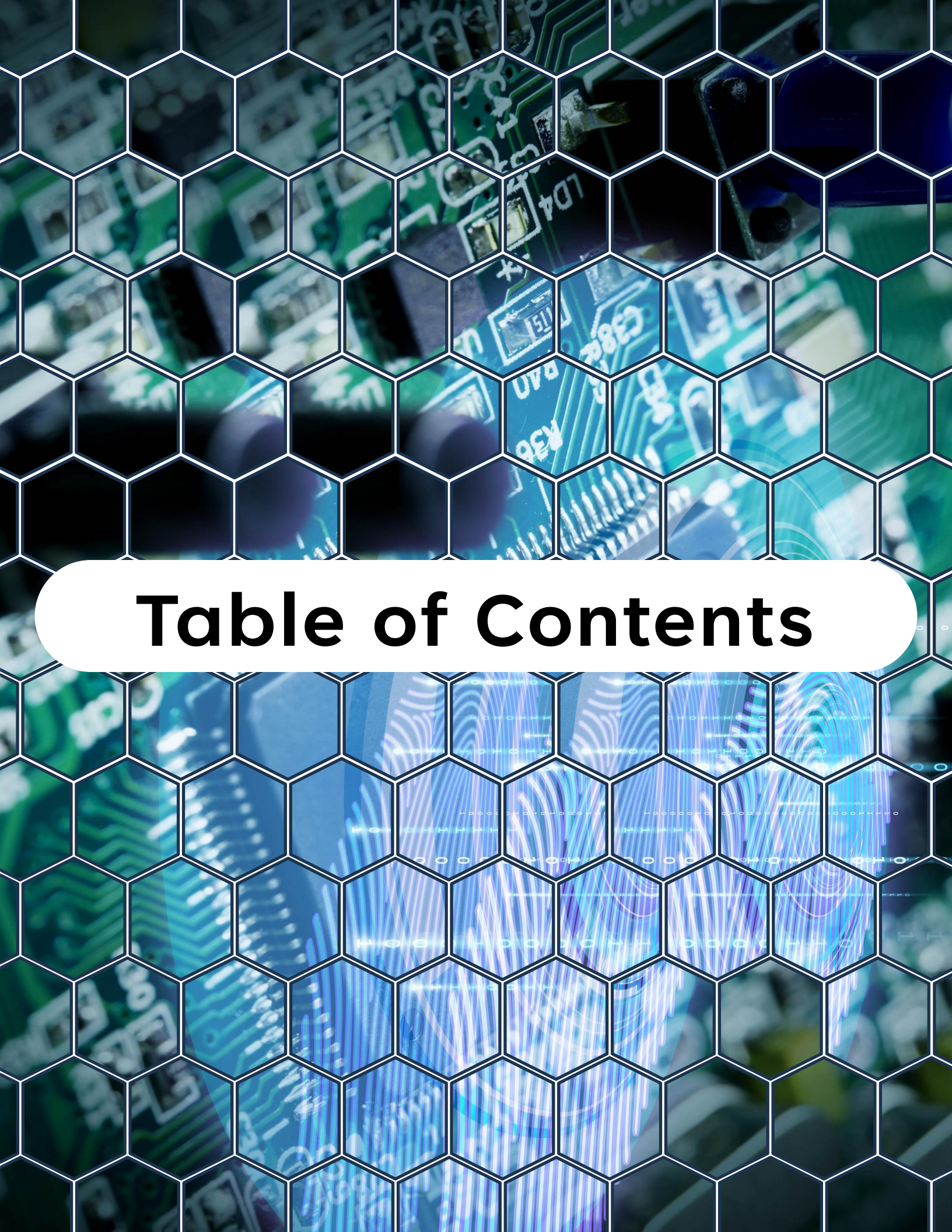
30-Day FreeRTOS Course for ESP32 Using ESP-IDF

(Day 16)

“Task Notifications”



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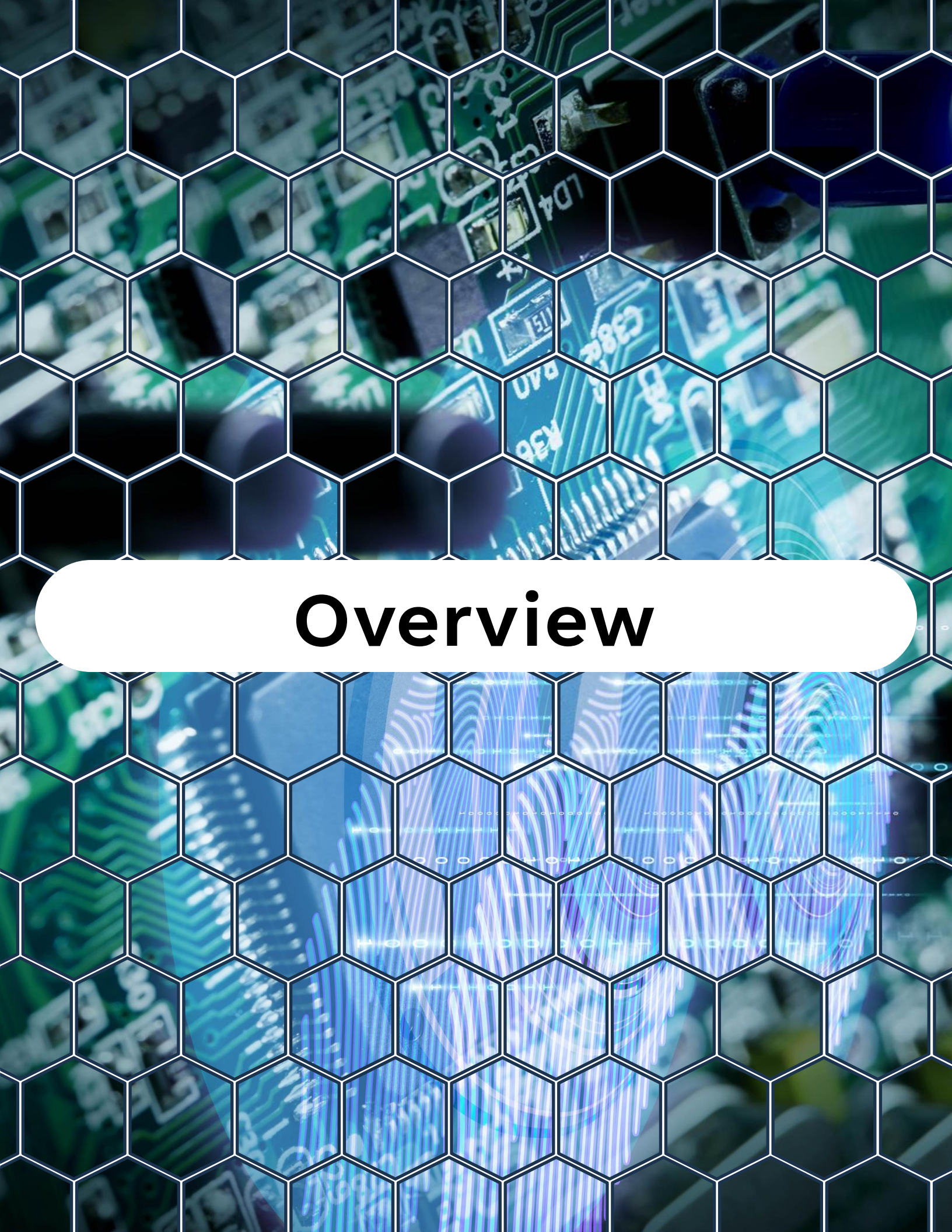

Overview

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On **Day 16**, you'll learn:

- What **task notifications** are in FreeRTOS
- How they compare to semaphores and event groups
- Different modes of task notification (overwrite, increment, bitwise, etc.)
- How to send notifications from tasks and ISRs
- A practical ESP32 example using **ISR → Task Notification**
- Best practices for task notification usage

By the end of this lesson, you'll know how to use task notifications as a lightweight, efficient way to synchronize tasks and ISRs.

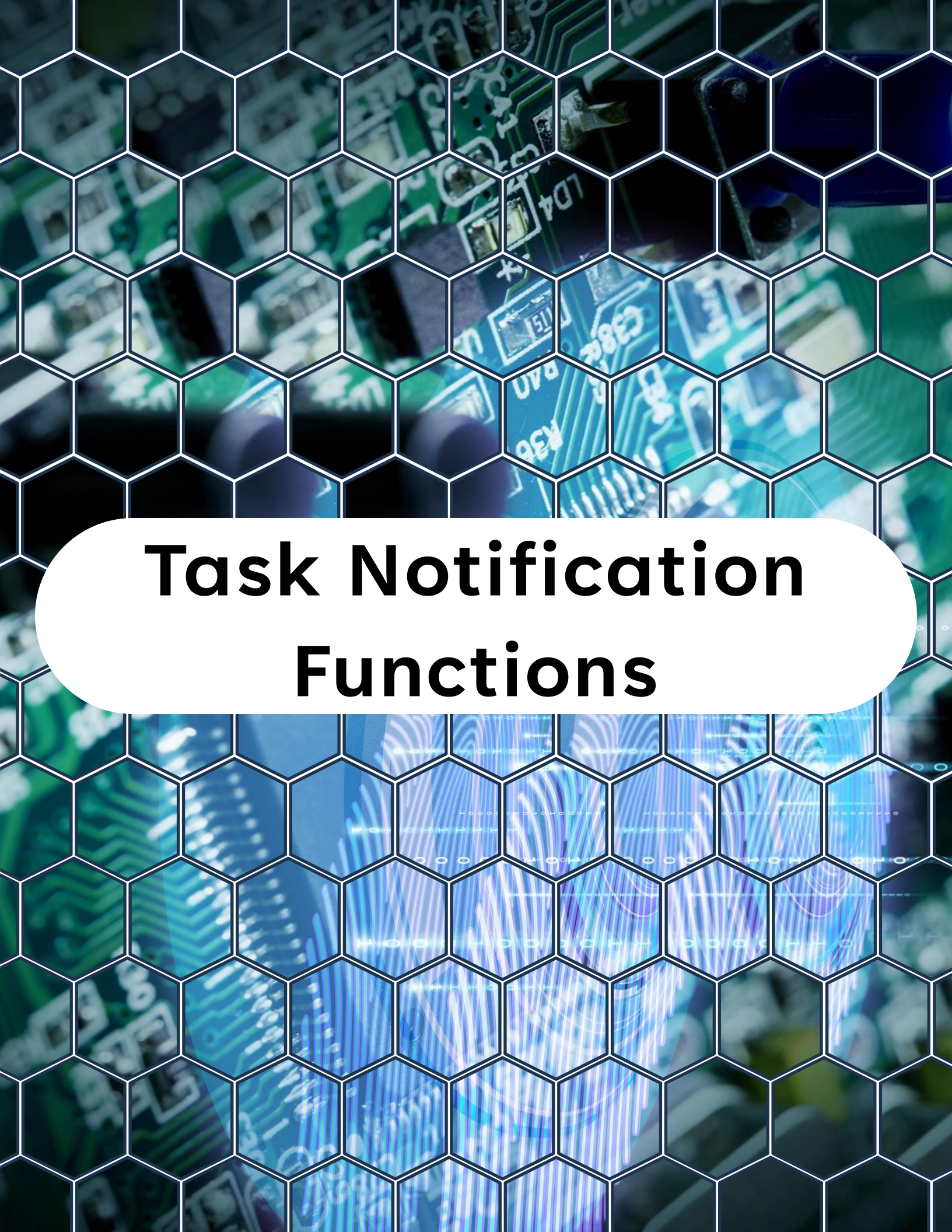

What Are Task Notifications?

What Are Task Notifications?

A **task notification** is a built-in synchronization mechanism in FreeRTOS.

- Each task has a **notification value** (like a private mailbox).
- Another task or ISR can **send a notification** to that task.
- The notified task can **read**, **clear**, or **wait** on its notification.

📌 Think of it as a **lightweight, one-to-one signaling system**, faster and more memory-efficient than semaphores or queues.



Task Notification Functions

Task Notification Functions

Sending Notifications

From another task:

```
xTaskNotify(TaskHandle_t xTaskToNotify,
            uint32_t ulValue,
            eNotifyAction eAction);
```

From ISR:

```
xTaskNotifyFromISR(TaskHandle_t xTaskToNotify,
                   uint32_t ulValue,
                   eNotifyAction eAction,
                   BaseType_t *pxHigherPriorityTaskWoken);
```

Receiving Notifications

```
xTaskNotifyWait(uint32_t ulBitsToClearOnEntry,
                uint32_t ulBitsToClearOnExit,
                uint32_t *pulNotificationValue,
                TickType_t xTicksToWait);
```


Notify Actions

Notify Actions

eNotifyAction defines how the notification value is updated:

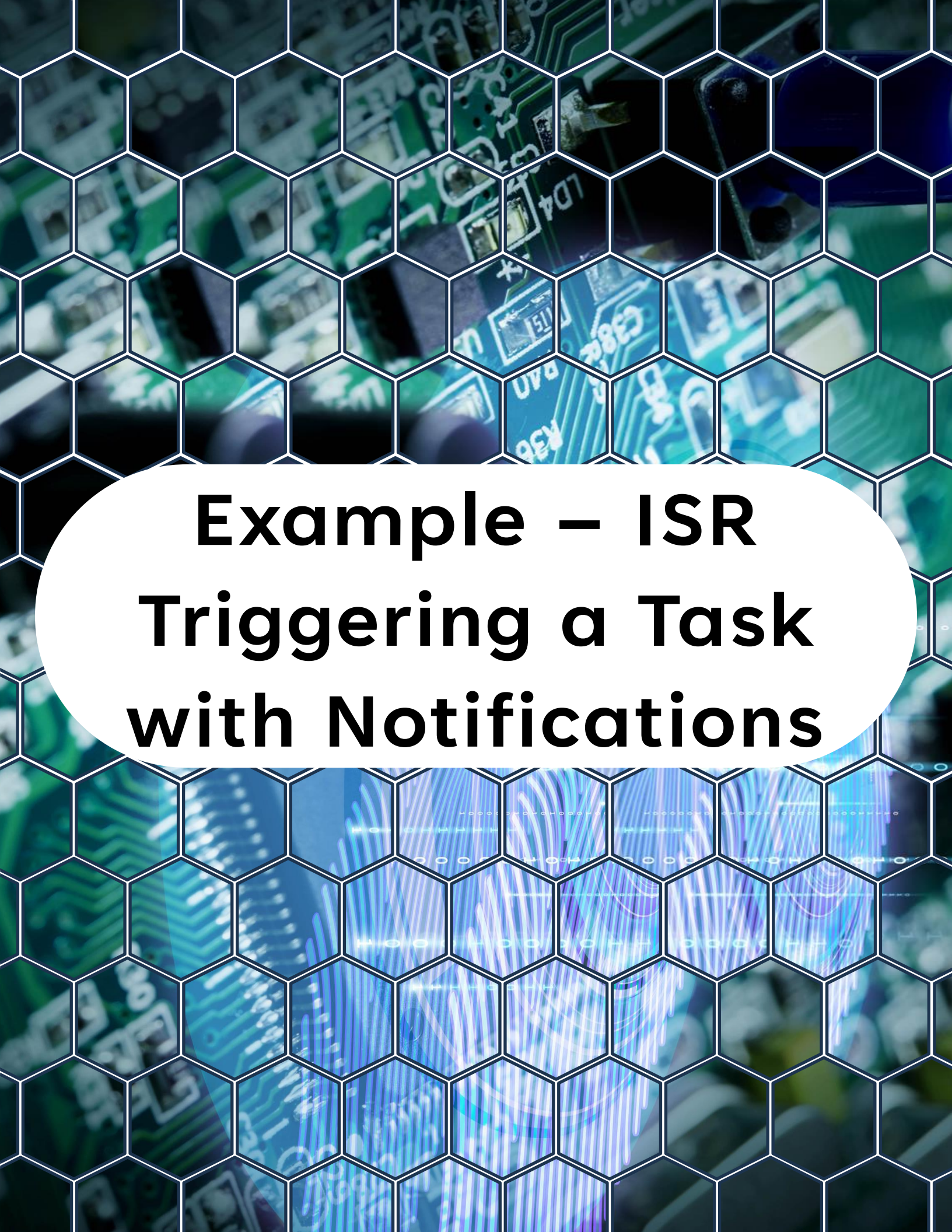
- `eNoAction` → just unblock the task (don't change value).
- `eSetBits` → set bits (bitwise OR).
- `eIncrement` → increment value.
- `eSetValueWithOverwrite` → overwrite with new value.
- `eSetValueWithoutOverwrite` → update only if no pending notification.


Why Use Task Notifications?

Why Use Task Notifications?

- **Lightweight** → No need to allocate memory (unlike queues).
- **Fast** → Directly signals the task, fewer instructions than semaphores.
- **Flexible** → Supports counting, bit flags, or simple signaling.
- **One-to-one** → Ideal for ISR → Task or Task → Task communication.

📌 If you only need to signal **one task**, task notifications are the most efficient option.



Example – ISR Triggering a Task with Notifications

Example – ISR Triggering a Task with Notifications

```c
#include <stdio.h>
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "driver/gpio.h"
#include "esp_log.h"

#define BUTTON_GPIO 0
#define TAG "DAY16"

// Task handle for the button task to allow
// notification from ISR
TaskHandle_t button_task_handle = NULL;

// ISR handler
static void IRAM_ATTR button_isr_handler(void *arg) {
    // Variable to check if a higher priority task
    // was woken by the notification
    BaseType_t xHigherPriorityTaskWoken = pdFALSE;

    // Notify the button task
    vTaskNotifyGiveFromISR(button_task_handle,
                           &xHigherPriorityTaskWoken);

    // Yield to higher priority task if necessary
    if (xHigherPriorityTaskWoken) {
        portYIELD_FROM_ISR();
    }
}

// Task waiting on notifications
void button_task(void *pvParameter) {
    while (1) {
```

Example – ISR Triggering a Task with Notifications

```
30 // Task waiting on notifications
31 void button_task(void *pvParameter) {
32     while (1) {
33         // Wait indefinitely for a notification
34         ulTaskNotifyTake(pdTRUE, portMAX_DELAY);
35         ESP_LOGI(TAG, "Button pressed! Task unblocked by ISR");
36     }
37 }
38
39 // Main application entry point
40 void app_main() {
41     // Configure button pin
42     gpio_config_t io_conf = {
43         .intr_type = GPIO_INTR_NEGEDGE,
44         .mode = GPIO_MODE_INPUT,
45         .pin_bit_mask = (1ULL << BUTTON_GPIO),
46         .pull_up_en = GPIO_PULLUP_ENABLE,
47         .pull_down_en = GPIO_PULLDOWN_DISABLE
48     };
49     gpio_config(&io_conf);
50
51     // Install ISR service and attach handler
52     gpio_install_isr_service(0);
53     gpio_isr_handler_add(BUTTON_GPIO, button_isr_handler, NULL);
54
55     // Create the button task
56     xTaskCreate(button_task, "ButtonTask", 2048, NULL, 10,
57                 &button_task_handle);
58 }
```

Example – ISR Triggering a Task with Notifications

Expected Behavior

- Press the button on GPIO0.
- The ISR runs and **gives a notification** to the waiting task.
- The button_task unblocks and logs a message.

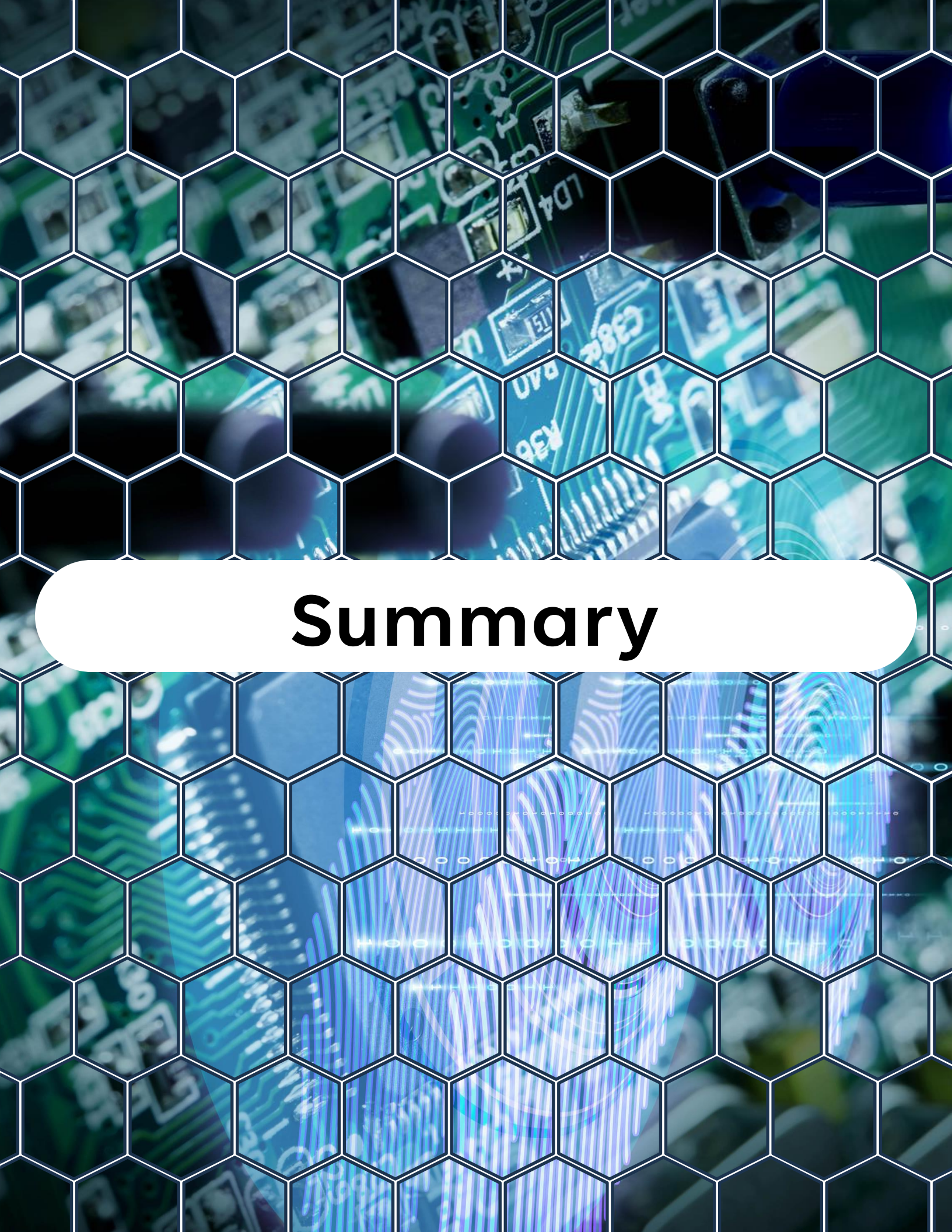
Output:

```
I (1234) DAY16: Button pressed! Task unblocked by ISR
```

Best Practices

Best Practices

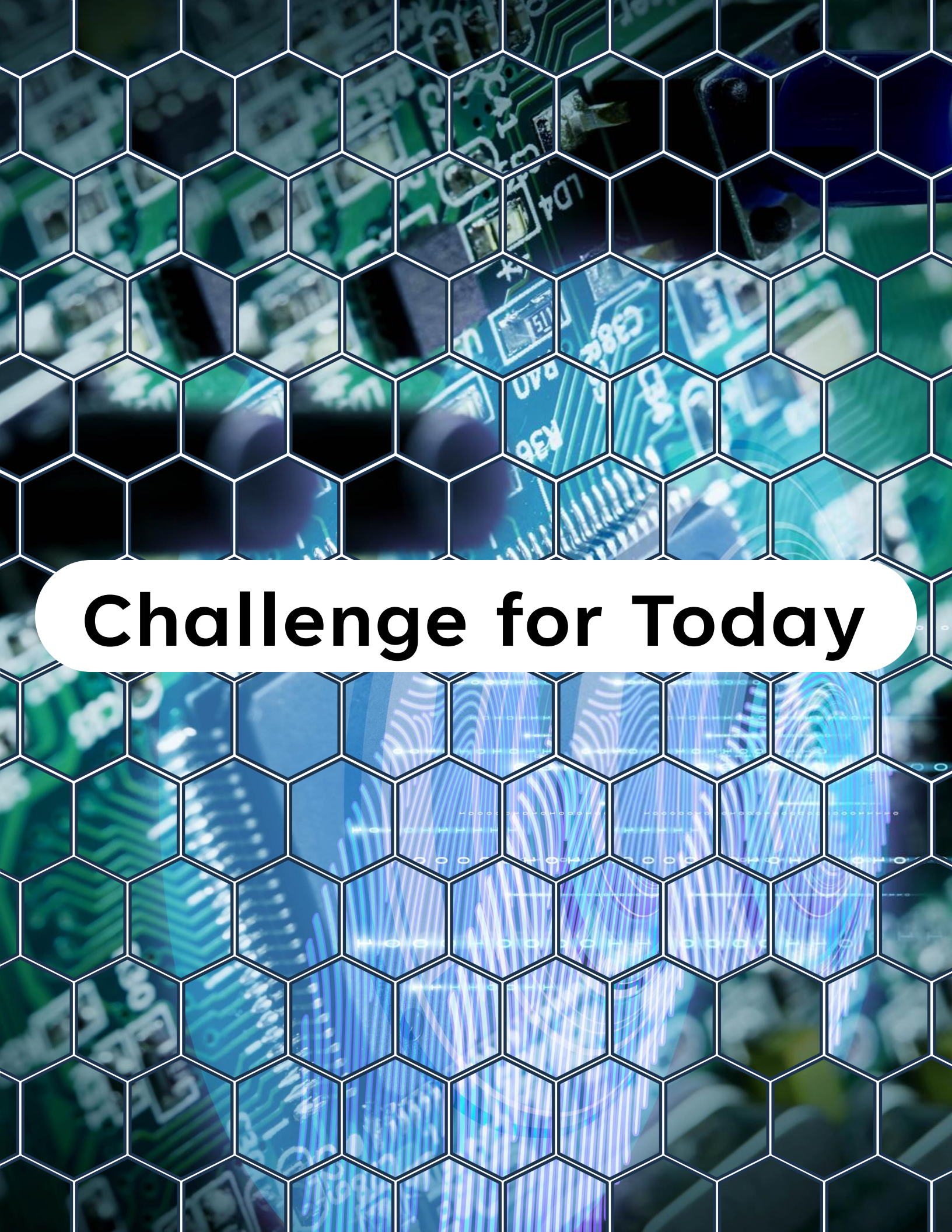
- ✅ Use **task notifications** for **one-to-one signaling** (ISR → Task, Task → Task).
- ✅ Use **queues** if you need to pass **data** or multiple messages.
- ✅ Use **event groups** if **multiple tasks** need to be synchronized.
- ✅ Use `ulTaskNotifyTake()` for counting events (like button presses).
- ✅ Keep ISRs short: only notify, don’t do logic inside.



Summary

Summary

- Task notifications are the **fastest, lightest** signaling mechanism in FreeRTOS.
- Each task has a built-in notification value.
- Notifications can increment, set bits, overwrite, or just signal.
- Perfect for **ISR** → **Task** signaling where performance matters.


Challenge for Today

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🛠️ Modify today's project by:

- Counting the number of button presses using `ulTaskNotifyTake(pdTRUE, ...)`.
- Print a running counter in the task (e.g., "Button pressed 5 times").
- Bonus: Use `xTaskNotify()` from another task to simulate button presses.

"Thanks for watching!

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